

AHHYUN YUH

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EDUCATION

Sep. 2022 ~ Present	University of Pennsylvania Department of the School of Engineering and Applied Science <i>PhD Candidate in Computer and Information Science</i> <i>Advisor: Prof. Insup Lee</i>	Philadelphia, PA
Mar. 2020 ~ Feb. 2022	Kyungpook National University Department of Electronic Engineering <i>Master of Science in Electronic Engineering</i> <i>Advisor: Prof. Soon Ju Kang</i>	Daegu, Korea
Mar. 2015 ~ Feb. 2020	Kyungpook National University Department of Electronic Engineering <i>Bachelor of Science in Electronic Engineering</i>	Daegu, Korea
Feb. 2018 ~ Aug. 2018	University of Ljubljana Faculty of Computer and Information Science <i>Exchange Student Program</i>	Ljubljana, Slovenia

PROJECTS

- **iCareChat: Voice-Based LLM Agent for Older Adults – Ongoing**

I am currently leading the iCareChat development, a privacy-preserving, HIPAA-compliant real-time voice chatbot app powered by large language models, designed to provide empathetic conversational support for older adults. The system uses a multi-agent framework and optimized audio pipelines and has been evaluated through cross-cultural studies in the US and Japan, as well as IRB-approved clinical experiments measuring mental health outcomes.

- **TrueDepth 3D Face Reconstruction for Graves' Disease Monitoring – Ongoing**

I am currently developing iOS app using Apple's TrueDepth camera and ARKit to collect and monitor facial morphology changes in Graves' disease patients. This project combines RGB-D data alignment, precise landmark correspondence, and integration into clinical workflows to enable privacy-conscious, at-home health monitoring.

- **iCareLoop: Social Isolation and Loneliness Detection**

I led the development of iCareLoop, an IoT-based sensing framework for detecting social isolation and loneliness among older adults through unobtrusive environmental monitoring. Using time-series analysis and behavioral pattern recognition, this system was validated in cross-cultural studies conducted in Japan and the US.

- **BlinkWise: Wearable Blink Dynamics Monitoring**

I contributed to building BlinkWise, a wearable system leveraging mmWave radar on top of ESP32 microcontroller to track blink patterns and infer real-time mental states. Advanced signal processing techniques such as FFT, beamforming, and adaptive filtering enabled low-power, high-accuracy blink detection.

- **Real-Time Sound Event Classification for Human Activity Recognition**

I developed an edge-based audio activity recognition system for continuous in-home monitoring, designed to preserve user privacy while providing accurate activity insights. This project involved designing non-invasive audio protocols and implementing a CNN-based classifier optimized for real-time inference on resource-constrained devices.

PUBLICATIONS

- “IoT-Based Loneliness Monitoring for Older Adults: Lessons from Japanese and US Communities”, *IEEE Access*, submitted, 2025
- “SMILE: Sensor Data-driven Detection and Proximate Cause Analysis for Loneliness in Older Adults”, *ACM Conference on Health, Inference, and Learning (ACM Health)*, submitted, 2025
- “Tracking Blink Dynamics and Mental States on Glasses”, *Proceedings of the ACM on Mobile Systems, Applications, and Services (MobiSys)*, accepted, 2025
- “Older Adults’ Perceptions and Attitudes Toward Passive Sensors to Measure Loneliness: A Qualitative Study”, *SAGE Open Aging*, 2025
- “Perceptions and Attitudes Toward Loneliness Detection using Passive Sensing Technologies in Older Adults”, *Innovation in Aging*, Vol. 8, Suppl. 1, 2024
- “Real-Time Prediction of Resident ADL Using Edge-Based Time-Series Ambient Sound Recognition”, *Sensors*, Vol. 24, No. 19, 2024
- “Exploring Effective Sensing Indicators of Loneliness for Elderly Community in US and Japan”, *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*, 2024
- “iCareLoop: Data Management System for Monitoring Gerontological Social Isolation and Loneliness”, *Proceedings of the 14th International Conference on Mobile Computing and Ubiquitous Networking (ICMU)*, 2023
- “Short: Integrated sensing platform for detecting social isolation and loneliness in the elderly community”, *IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2023
- “iCareLoop: Closed-Loop Sensing and Intervention for Gerontological Social Isolation and Loneliness”, *Proceedings of the ACM/IEEE 14th International Conference on Cyber-Physical Systems (ICCPS)*, 2023
- “Real-Time Sound Event Classification for Human Activity of Daily Living using Deep Neural Network”, *Proceedings of the 14th IEEE International Conference on Internet of Things (iThings)*, Melbourne, 2021
- “Self-Organized Sensor Fusion between IoT Devices for Resident Activity Recognition and Evaluation System”, *Proceedings of the Institute of Embedded Engineering of Korea Conference*, in Korean, 2020

WORK EXPERIENCE

May. 2020 ~ Feb. 2022	MATLAB Student Ambassador https://www.facebook.com/groups/KNUMATLAB ▪ Proceed seminars and events regarding MATLAB ▪ Manage Kyungpook National University Facebook and Instagram pages and post works ▪ Student mentor regarding MATLAB and Simulink	MathWorks Korea
Mar. 2020 ~ Feb. 2022	Center of Self-Organizing Software Researcher http://www.csosp.org/ ▪ Participate in Self-Organized Ambient Assisted Living (SAAL) Project	Kyungpook National University
Sep. 2019 ~ Feb. 2022	Real-Time System Lab Researcher https://sites.google.com/view/knu-rtlab/ ▪ Research student at Real-Time System Lab	Kyungpook National University
Mar. 2018 ~ Feb. 2019	RLS Internship https://www.rls.si/ ▪ Assistant for engineers at SPS Germany 2018 ▪ Korean translation in general technical news, descriptions and datasheets for RLS website	Vodice, Slovenia

- Advertisement and sales plan targeting Korean market

AWARDS AND HONORS

Nov. 2021	2021 KNU-EE Research Congress <i>2nd place</i> Real-Time Sound Event Classification for Human Activity of Daily Living using Deep Neural Network	Kyungpook National University
Mar. 2019	Startup Idea Competition <i>2nd place in Startup Idea and Pitching Competition</i> Proposition of a new business model regarding E-commerce, supply chain and virtual fabric printing	Gyeongsangbukdo Creative Content Agency
Nov. 2015	Intelligent Robot Tournament <i>1st place in pick-and-place Robot Tournament</i> Design, model and control of robots to pick and place figures and ball bearings to save the contest mission	Kyungpook National University